Dimensional Analysis and Significant Figures

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https://github.com/rajatkalia1982/projects

https://sourceforge.net/u/rajatkalia/profile

https://www.youtube.com/@rajatkaliaeducation

https://www.lulu.com/spotlight/rajatkalia11

https://amazon.com/author/rajatkalia

https://rk-e.blogspot.in/

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Part I

Dimensional Analysis

Problem Set

Chapter 1

JEE Mains Problems

1.1 Intext Problems

Q1: The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = Electric Field) is $M^a L^b T^c$. The value of $2a - b + c = 0$ is _________.

[JEE Mains 2nd April 2026 Shift 1]
A) 0
B) 1
C) -1
D) 2
{Solution : https://www.youtube.com/watch?v=CAgTYRH7-P8 }

1.2 Full Syllabus Problems

Q1: The position of an object having mass 0.1 kg as a function of time t is given as $\overrightarrow{r} = 10t^2\hat{i} + 5t^3\hat{j}$ m. At t = 1 s, which of the following statements is correct ?

[JEE Mains 2nd April 2026 Shift 1]
A The linear momentum $\overrightarrow{p} = (2\hat{i} + 1.5\hat{j})kg.m/s.$
B The force acting on the object $\overrightarrow{F} = \left(2\hat{i} + 3\hat{j}\right) N.$
C The angular momentum of the object about it's origin $\overrightarrow{L} = 15\hat{k}\ J.s.$
D The torque acting on the object about it's origin $T = 20\hat{k}\ N.m.$
Choose the correct answer from the options given below :
(a) A, B and C only.
(b) B, C and D only.
(c) A, C and D only.
(d) A, B and D only.
{Solution : https://www.youtube.com/watch?v=DvIERvObc1E}

Part II

Significant Figures

Problem Set

Chapter 2

JEE Mains Problems

Q1 : The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measurement of Young's Modulus is $\alpha \times 10^9\ N/m^2$. The value of α is _________.

(Ignore the contribution of load to Young's Modulus error calculation)

[JEE Mains 2nd April 2026 Shift 1]

A) 1.3

B) 1.65

C) 0.13

D) 0.25

{Solution : https://www.youtube.com/watch?v=JCaMvvfacDw}